

Curriculum Vitae / Resume

Avinash Gyawali

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SUMMARY

First-year master's student in Computer Vision with a research focus on efficient multimodal large language models, 3D reconstruction and streaming video understanding,

EDUCATION

University of Central Florida (UCF) <i>M.Sc. in Computer Vision</i>	2025 - 2027 GPA: 3.8/4.0
New York University Abu Dhabi <i>Bachelor of Computer Science</i>	2021 - 2025 GPA: 3.78/4.0

PUBLICATIONS

[PRE-PRINT] Segmented Visual Geometry Grounded Transformer <i>Avinash Gyawali, Yushra Ahmed</i>	August 2025
[ACCEPTED] A Custom Haptic Syringe to Improve Virtual Reality Local Anesthesia Surgical Training <i>Avinash Gyawali, Mohamad Eid, Marci Levine, Elizabeth C McAlpin</i>	November 2024

EXPERIENCE

Center for Research in Computer Vision , UCF <i>Computer Vision Research Assistant</i>	Orlando , USA August 2025 – Nov 2025
- Designed and implemented a knowledge distillation pipeline to transfer zero-shot image classification capabilities from CLIP's ViT-based visual encoder (e.g., FastViT) to a lightweight ResNet-50 backbone. - Integrated adapter layers, LoRA, and visual prompt tuning to align self-attention-based representations with convolutional feature hierarchies during distillation, achieving 5% improvement in zero-shot accuracy and 10× reduction in model size/inference	
Applied Multimedia Lab, NYUAD  <i>Research Assistant</i>	Abu Dhabi, UAE August 2024 – May 2025
- Developed a emotion classification model and optimized a transformer-based ATCNet model to classify five emotional states from EEG signals, improving model accuracy from an initial 68% to a peak of 74.45% through systematic hyperparameter tuning - Engineered a deep learning pipeline for emotion recognition by implementing an ATCNet architecture that uses convolutional layers, multi-head self-attention, and temporal convolutional networks to effectively extract spatiotemporal features from raw EEG data	
Empathic Computing Lab <i>HCI Research Intern, Advised by Dr. Mark Billinghurst</i>	Adelaide, Australia June 2024 – December 2024
- Fine-tuned and deployed a Llama-8B LLM using parameter-efficient QLoRA to accurately classify deictic language from live speech, enabling real-time, low-latency visual cues in a collaborative VR application. - Engineered an end-to-end machine learning pipeline in Unity that integrated OpenAI's Whisper for speech-to-text with the custom-tuned LLM to seamlessly translate conversational intent into visual aids.	

AWARDS

NYUAD Summer Fellowship: Received grant to pursue research in Australia(\$5000)
Microsoft Hololens Hackathon 🏆: 1st place among 10 universities in the UAE (\$8000 Prize Money) 
Meta AWS WebXR Hackathon 🏆: 2nd place among 15 teams 
Microsoft Imagine Cup 🏆: Top 10 finalists from Asia (Project Name: EDVR)
Tie Startup Competition 🏆: Top 10 finalists from the Middle East
NASA Space Apps Regional 🏆: Top 2 among 20+ teams

SKILLS

Programming languages: Python, C++, C#, SQL
ML Tools: PyTorch, OpenMMLab , HuggingFace Transformers, CLIP, OpenCV, LoRA , Jupyter, Pandas, NumPy , DINO
Cloud Frameworks: AWS , Azure ML
Human Languages: English, Nepali